

Lighthouse prompt

A compact request with large hidden state

Design a low-power **64-bit RISC-V-based compute subsystem** for an **XR Bench-class real-time mobile XR workload**, decide whether it is a **vector-capable CPU, accelerator, or SoC block**, meet a **3 W TDP target in a 3 nm-class LP mobile process**, and produce a **design-space report with evidence and rejected alternatives**.



Architecture obligations

What the prompt immediately forces an architect to define

XR Bench workload

RT-MTMM, streams, deadlines

ISA and ABI contract

RISC-V baseline, extensions, compatibility

CPU / accelerator / SoC block

partitioning, vector support, interfaces

Memory and data movement

locality, hierarchy, interconnect energy

3 W power envelope

activity, thermal, energy/QoE tradeoff

Compiler/runtime stack

codegen, libraries, deployment

3 nm-class physical constraints

process, timing, leakage, signoff

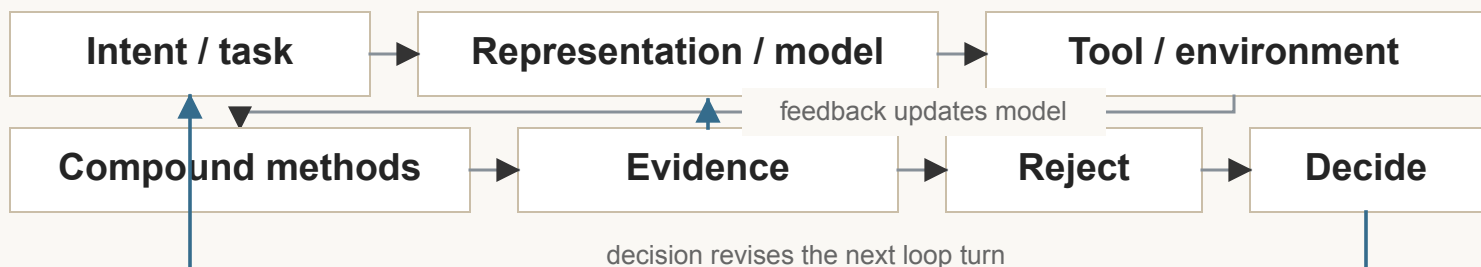
Reliability and verification

correctness, corner cases, rejection



Design-loop requirements

One design turn: represent, act, gather evidence, reject, decide, revise



Boundary condition

Prompt-to-loop design with evidence, rejection, and judgment, not prompt-to-chip automation.